B.E. ELECTRICAL ENGINEERING

Second Year Second Semester Examination, 2026

DIGITAL SIGNAL PROCESSING

PART II - COMPREHENSIVE SOLUTIONS

Prepared meticulously in the pedagogical style of SGM Sir and BJB Sir

Featuring exhaustive step-by-step mathematical derivations, complete block diagrams, and strict adherence to university exam formatting.

Includes ALL primary questions AND all "OR" alternate questions.

Complete Answer Script

Contents

Question 1: Fourier Series to Sine/Cosine Coefficients

Problem Statement

Show in detail how the Fourier series of a periodic signal can be computed in terms of sine coefficients and cosine coefficients. How are the amplitude and phase of the nth harmonic related to these sine and cosine coefficients?

1. Derivation of the Trigonometric Fourier Series

Let $x(t)$ be a continuous-time periodic signal with a fundamental period T_0. The fundamental angular frequency is defined as:

$$\omega_0 = \frac{2\pi}{T_0} \tag{1}$$

According to the foundational theorem of Fourier analysis, any well-behaved periodic signal can be expressed as an infinite sum of complex exponentials (the Exponential Fourier Series):

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \tag{2}$$

where the complex Fourier coefficients c_n are computed using the synthesis equation:

$$c_n = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jn\omega_0 t}dt \tag{3}$$

To convert this complex exponential series into the trigonometric series containing sine and cosine terms, we apply **Euler's Identity**:

$$e^{j\theta} = \cos(\theta) + j\sin(\theta) \tag{4}$$

$$e^{-j\theta} = \cos(\theta) - j\sin(\theta) \tag{5}$$

We can separate the infinite summation into three distinct parts: the DC component ($n = 0$), the positive frequency harmonics ($n > 0$), and the negative frequency harmonics ($n < 0$).

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n e^{jn\omega_0 t} + \sum_{n=-\infty}^{-1} c_n e^{jn\omega_0 t} \tag{6}$$

Let us re-index the third term by replacing n with $-n$:

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n e^{jn\omega_0 t} + \sum_{n=1}^{\infty} c_{-n} e^{-jn\omega_0 t} \tag{7}$$

Combining the summations:

$$x(t) = c_0 + \sum_{n=1}^{\infty} \left[c_n e^{jn\omega_0 t} + c_{-n} e^{-jn\omega_0 t}\right] \tag{8}$$

Now, substituting Euler's identities into the exponential terms:

$$e^{jn\omega_0 t} = \cos(n\omega_0 t) + j\sin(n\omega_0 t) \tag{9}$$

$$e^{-jn\omega_0 t} = \cos(n\omega_0 t) - j\sin(n\omega_0 t) \tag{10}$$

Substituting these back into the series equation gives:

$$x(t) = c_0 + \sum_{n=1}^{\infty} \left[c_n(\cos(n\omega_0 t) + j\sin(n\omega_0 t)) + c_{-n}(\cos(n\omega_0 t) - j\sin(n\omega_0 t))\right] \tag{11}$$

Regrouping the terms by factoring out the cosine and sine components:

$$x(t) = c_0 + \sum_{n=1}^{\infty} \left[(c_n + c_{-n})\cos(n\omega_0 t) + j(c_n - c_{-n})\sin(n\omega_0 t)\right] \tag{12}$$

We now formally define the trigonometric Fourier coefficients (the sine and cosine coefficients):

$$a_0 = c_0 \tag{13}$$

$$a_n = c_n + c_{-n} \tag{14}$$

$$b_n = j(c_n - c_{-n}) \tag{15}$$

This allows us to write the definitive **Trigonometric Fourier Series**:

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n\cos(n\omega_0 t) + b_n\sin(n\omega_0 t)\right] \tag{16}$$

2. Computation of the Coefficients (a_n and b_n)

From the definition of c_n:

$$c_n = \frac{1}{T_0}\int_0^{T_0} x(t)(\cos(n\omega_0 t) - j\sin(n\omega_0 t))dt \tag{17}$$

Since $x(t)$ is a real-valued signal, c_{-n} is the complex conjugate of c_n ($c_{-n} = c_n^*$). Therefore:

$$a_n = c_n + c_n^* = 2\ \mathrm{Re}\{c_n\} = \frac{2}{T_0}\int_0^{T_0} x(t)\cos(n\omega_0 t)dt \tag{18}$$

$$b_n = j(c_n - c_n^*) = j(j2\ \mathrm{Im}\{c_n\}) = -2\ \mathrm{Im}\{c_n\} = \frac{2}{T_0}\int_0^{T_0} x(t)\sin(n\omega_0 t)dt \tag{19}$$

And for the DC term:

$$a_0 = \frac{1}{T_0}\int_0^{T_0} x(t)dt \tag{20}$$

3. Amplitude and Phase of the nth Harmonic

The n-th harmonic consists of the terms:

$$H_n(t) = a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \tag{21}$$

We wish to express this as a single cosine wave with a specific amplitude A_n and phase shift ϕ_n:

$$H_n(t) = A_n \cos(n\omega_0 t + \phi_n) \tag{22}$$

Using the trigonometric identity for the cosine of a sum:

$$A_n \cos(n\omega_0 t + \phi_n) = A_n \cos(n\omega_0 t)\cos(\phi_n) - A_n \sin(n\omega_0 t)\sin(\phi_n) \tag{23}$$

Comparing the coefficients of $\cos(n\omega_0 t)$ and $\sin(n\omega_0 t)$ from both equations, we establish:

$$a_n = A_n \cos(\phi_n) \tag{24}$$

$$b_n = -A_n \sin(\phi_n) \tag{25}$$

To find the Amplitude (A_n): We square both equations and add them together:

$$a_n^2 + b_n^2 = A_n^2 \cos^2(\phi_n) + A_n^2 \sin^2(\phi_n) = A_n^2(\cos^2(\phi_n) + \sin^2(\phi_n)) = A_n^2 \tag{26}$$

Thus, the magnitude of the amplitude is:

$$\boxed{A_n = \sqrt{a_n^2 + b_n^2} \qquad (27)}$$

To find the Phase (ϕ_n): We divide the equation for b_n by the equation for a_n:

$$\frac{b_n}{a_n} = \frac{-A_n \sin(\phi_n)}{A_n \cos(\phi_n)} = -\tan(\phi_n) \tag{28}$$

Thus, the phase is given by:

$$\boxed{\phi_n = -\tan^{-1}\left(\frac{b_n}{a_n}\right) \qquad (29)}$$

Question 1 (OR): Inverse Discrete Fourier Transform (IDFT)

Problem Statement

Using the DFT coefficients $X_k = [20, -4+4j, -4, -4-4j]$, compute the inverse discrete Fourier transform to obtain the time domain sequence $x_n, n = 0, 1, 2, 3$.

1. Mathematical Formulation of the IDFT

The Inverse Discrete Fourier Transform (IDFT) is defined by the formula:

$$x_n = \frac{1}{N}\sum_{k=0}^{N-1} X_k W_N^{-kn} \quad \text{for } n = 0, 1, \ldots, N-1 \tag{30}$$

where W_N is the twiddle factor, defined as:

$$W_N = e^{-j\frac{2\pi}{N}} \tag{31}$$

For this problem, the length of the sequence is $N = 4$. Therefore:

$$W_4 = e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}} = \cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right) = -j \tag{32}$$

The inverse twiddle factor required for the IDFT is:

$$W_4^{-1} = e^{j\frac{\pi}{2}} = j \tag{33}$$

We are given the discrete frequency sequence:

- $X_0 = 20$
- $X_1 = -4 + 4j$
- $X_2 = -4$
- $X_3 = -4 - 4j$

2. Step-by-Step Calculation

We will evaluate the summation $x_n = \frac{1}{4}\sum_{k=0}^{3} X_k(W_4^{-1})^{kn} = \frac{1}{4}\sum_{k=0}^{3} X_k(j)^{kn}$ for each value of n.

Calculation for $n = 0$:

$$x_0 = \frac{1}{4}\left[X_0(j)^0 + X_1(j)^0 + X_2(j)^0 + X_3(j)^0\right] \tag{34}$$

$$x_0 = \frac{1}{4}\left[20 + (-4 + 4j) + (-4) + (-4 - 4j)\right] \tag{35}$$

Gathering the real parts and imaginary parts:

$$x_0 = \frac{1}{4}\left[20 - 4 - 4 - 4 + j(4 - 4)\right] = \frac{1}{4}[8 + 0j] = 2 \tag{36}$$

Calculation for $n = 1$:

$$x_1 = \frac{1}{4}\left[X_0(j)^0 + X_1(j)^1 + X_2(j)^2 + X_3(j)^3\right] \tag{37}$$

Recall the powers of j: $j^1 = j$, $j^2 = -1$, $j^3 = -j$.

$$x_1 = \frac{1}{4}\left[20(1) + (-4+4j)(j) + (-4)(-1) + (-4-4j)(-j)\right] \tag{38}$$

Let us carefully expand the complex multiplications:

- $(-4+4j)(j) = -4j + 4j^2 = -4j - 4$
- $(-4)(-1) = 4$
- $(-4-4j)(-j) = 4j + 4j^2 = 4j - 4$

Substituting these expansions back into the summation:

$$x_1 = \frac{1}{4}\left[20 - 4j - 4 + 4 + 4j - 4\right] \tag{39}$$

The imaginary terms cancel out $(-4j + 4j = 0)$. Summing the real terms:

$$x_1 = \frac{1}{4}\left[20 - 4 + 4 - 4\right] = \frac{16}{4} = 4 \tag{40}$$

Calculation for $n = 2$:

$$x_2 = \frac{1}{4}\left[X_0(j)^0 + X_1(j)^2 + X_2(j)^4 + X_3(j)^6\right] \tag{41}$$

Powers of j: $j^2 = -1$, $j^4 = 1$, $j^6 = -1$.

$$x_2 = \frac{1}{4}\left[20(1) + (-4+4j)(-1) + (-4)(1) + (-4-4j)(-1)\right] \tag{42}$$

$$x_2 = \frac{1}{4}\left[20 + 4 - 4j - 4 + 4 + 4j\right] \tag{43}$$

The imaginary terms cancel out ($-4j + 4j = 0$). Summing the real terms:

$$x_2 = \frac{1}{4}\left[20 + 4 - 4 + 4\right] = \frac{24}{4} = 6 \tag{44}$$

Calculation for $n = 3$:

$$x_3 = \frac{1}{4}\left[X_0(j)^0 + X_1(j)^3 + X_2(j)^6 + X_3(j)^9\right] \tag{45}$$

Powers of j: $j^3 = -j$, $j^6 = -1$, $j^9 = j$.

$$x_3 = \frac{1}{4}\left[20(1) + (-4+4j)(-j) + (-4)(-1) + (-4-4j)(j)\right] \tag{46}$$

Expanding the complex multiplications:

- $(-4+4j)(-j) = 4j - 4j^2 = 4j + 4$
- $(-4)(-1) = 4$
- $(-4-4j)(j) = -4j - 4j^2 = -4j + 4$

$$x_3 = \frac{1}{4}\left[20 + 4j + 4 + 4 - 4j + 4\right] \tag{47}$$

The imaginary terms cancel. Summing the real terms:

$$x_3 = \frac{1}{4}[20 + 4 + 4 + 4] = \frac{32}{4} = 8 \tag{48}$$

Final Result: The discrete-time domain sequence is precisely recovered without any imaginary residues:

$$\boxed{x_n = \{2, 4, 6, 8\}} \tag{49}$$

Question 2(a): Circular Symmetry & Modulo Operation

Problem Statement

What is circular symmetry of a sequence? What is the importance of "modulo" operation in it? Explain with a suitable example.

1. Definition of Circular Symmetry

In the domain of the Discrete Fourier Transform (DFT), signals are inherently treated as periodic with a period of N. Due to this strict N-point periodicity constraint, the standard linear time-reversal $x[-n]$ does not map correctly within the finite duration boundary $0 \leq n \leq N-1$.

To resolve this, we define **Circular Time Reversal** (or Circular Folding). A sequence $x[n]$ exhibits circular symmetry if it is identical to its circularly folded counterpart. The circularly folded sequence is denoted mathematically as $x[(-n)]_N$.

2. The Importance of the "Modulo" Operation

The modulo operation, denoted as $\mod N$ or $((\cdot))_N$, ensures that any time index falls strictly within the primary range $0 \leq n \leq N-1$. It behaves like a clock face with N hours.

When we reverse the time axis for a finite N-point sequence to find $x[-n]$, the indices become negative (e.g., $n = -1, -2$). The modulo operation forces these negative indices to wrap around to the positive end of the sequence:

$$(-n) \mod N = N - n \quad \text{for } 0 < n < N \tag{50}$$

For $n = 0$, the modulo remains 0. Therefore, the circular time reversal is defined as:

$$x[(-n)]_N = \begin{cases} x[0] & n = 0 \\ x[N-n] & 1 \leq n \leq N-1 \end{cases} \tag{51}$$

The modulo operation is the mathematical engine that enforces the periodic boundary conditions required by the DFT.

3. Suitable Example

Let us take an $N = 4$ point sequence:

$$x[n] = \{x[0], x[1], x[2], x[3]\} = \{1, 2, 3, 4\} \tag{52}$$

We wish to compute its circularly folded version, $x[(-n)]_4$:

- For $n = 0$: Index is $0 \mod 4 = 0 \implies x[0] = 1$
- For $n = 1$: Index is $-1 \mod 4 = 3 \implies x[3] = 4$
- For $n = 2$: Index is $-2 \mod 4 = 2 \implies x[2] = 3$
- For $n = 3$: Index is $-3 \mod 4 = 1 \implies x[1] = 2$

Thus, the circularly folded sequence is:

$$x[(-n)]_4 = \{1, 4, 3, 2\} \tag{53}$$

Notice how the first element $x[0]$ remains anchored, while the rest of the sequence is horizontally flipped as if wrapped around a cylinder. If $x[n]$ was initially $\{1, 4, 2, 4\}$, its circularly folded version would also be $\{1, 4, 2, 4\}$. Such a sequence perfectly satisfies the condition $x[n] = x[(-n)]_N$ and is said to possess **Even Circular Symmetry**.

Question 2(b): Impulse Sequence Symmetry of Causal FIR Filters

Problem Statement

Derive the symmetry property of the impulse sequence of a causal FIR filter. Then derive the frequency response of this causal FIR digital filter.

1. Derivation of the Symmetry Property

An essential feature often required in Digital Signal Processing is **exact linear phase**, which prevents phase distortion (i.e., all frequency components are delayed by the exact same amount of time).

The frequency response of a generic causal FIR filter of length M (spanning $n = 0$ to $M - 1$) is:

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n]e^{-j\omega n} \tag{54}$$

We can express this complex frequency response in polar form:

$$H(e^{j\omega}) = \pm|H(e^{j\omega})|e^{j\theta(\omega)} \tag{55}$$

For the filter to exhibit strict linear phase, the phase angle must be a linear function of frequency ω:

$$\theta(\omega) = -\alpha\omega \tag{56}$$

where α represents a constant group delay ($\alpha = \frac{M-1}{2}$).

Substituting this phase back into the equation and taking the ratio of the imaginary part to the real part of $H(e^{j\omega})$:

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n]\cos(\omega n) - j\sum_{n=0}^{M-1} h[n]\sin(\omega n) \tag{57}$$

$$\tan(\theta(\omega)) = \frac{-\sum_{n=0}^{M-1} h[n]\sin(\omega n)}{\sum_{n=0}^{M-1} h[n]\cos(\omega n)} \tag{58}$$

Substitute the linear phase requirement $\theta(\omega) = -\alpha\omega$:

$$\frac{-\sin(\alpha\omega)}{\cos(\alpha\omega)} = \frac{-\sum_{n=0}^{M-1} h[n]\sin(\omega n)}{\sum_{n=0}^{M-1} h[n]\cos(\omega n)} \tag{59}$$

Cross-multiplying gives:

$$\sum_{n=0}^{M-1} h[n]\sin(\omega n)\cos(\alpha\omega) = \sum_{n=0}^{M-1} h[n]\cos(\omega n)\sin(\alpha\omega) \tag{60}$$

Bring all terms to one side:

$$\sum_{n=0}^{M-1} h[n]\left[\sin(\omega n)\cos(\alpha\omega) - \cos(\omega n)\sin(\alpha\omega)\right] = 0 \tag{61}$$

Applying the trigonometric identity $\sin(A-B) = \sin A\cos B - \cos A\sin B$:

$$\sum_{n=0}^{M-1} h[n]\sin(\omega(n-\alpha)) = 0 \tag{62}$$

For this infinite sum to identically evaluate to zero for *all* possible values of ω, the individual terms must cancel out symmetrically. This happens if and only if:

$$\alpha = \frac{M-1}{2} \tag{63}$$

and the sequence coefficients are perfectly symmetric about this center point α:

$$\boxed{h[n] = h[M-1-n] \quad \text{for } 0 \leq n \leq M-1} \tag{64}$$

This proves that a causal FIR filter will exhibit exactly linear phase if and only if its impulse response $h[n]$ possesses **positive symmetry** around its center $(M-1)/2$. (Note: negative symmetry $h[n] = -h[M-1-n]$ yields a constant phase offset of $\pi/2$, which is utilized for differentiators and Hilbert transformers).

2. Derivation of the Frequency Response (Assuming M is Odd)

Let us assume the filter length M is odd. The center of symmetry is an integer index $n = (M-1)/2$. The standard DTFT is:

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n]e^{-j\omega n} \tag{65}$$

We can split this summation into two symmetric halves, isolating the center term:

$$H(e^{j\omega}) = \sum_{n=0}^{(M-3)/2} h[n]e^{-j\omega n} + h\left[\frac{M-1}{2}\right] e^{-j\omega\frac{M-1}{2}} + \sum_{n=(M+1)/2}^{M-1} h[n]e^{-j\omega n} \tag{66}$$

In the third summation (the upper half), let us perform a change of variable. Let $m = M-1-n$. As n ranges from $(M+1)/2$ up to $M-1$, the new variable m will range from $(M-3)/2$ down to 0.

$$\sum_{n=(M+1)/2}^{M-1} h[n]e^{-j\omega n} = \sum_{m=0}^{(M-3)/2} h[M-1-m]e^{-j\omega(M-1-m)} \tag{67}$$

Substitute the derived symmetry property $h[M-1-m] = h[m]$:

$$= \sum_{m=0}^{(M-3)/2} h[m]e^{-j\omega(M-1)}e^{j\omega m} \tag{68}$$

Now, re-insert this into the main $H(e^{j\omega})$ equation and factor out the common phase term $e^{-j\omega(M-1)/2}$ from all parts:

$$H(e^{j\omega}) = e^{-j\omega\frac{M-1}{2}} \left[\sum_{n=0}^{(M-3)/2} h[n] e^{j\omega\left(\frac{M-1}{2}-n\right)} + h\left[\frac{M-1}{2}\right] + \sum_{n=0}^{(M-3)/2} h[n] e^{-j\omega\left(\frac{M-1}{2}-n\right)} \right] \tag{69}$$

We group the exponentials with matching coefficients $h[n]$ together:

$$H(e^{j\omega}) = e^{-j\omega\frac{M-1}{2}} \left[h\left[\frac{M-1}{2}\right] + \sum_{n=0}^{(M-3)/2} h[n] \left(e^{j\omega\left(\frac{M-1}{2}-n\right)} + e^{-j\omega\left(\frac{M-1}{2}-n\right)} \right) \right] \tag{70}$$

Using Euler's identity $e^{j\theta} + e^{-j\theta} = 2\cos(\theta)$:

$$H(e^{j\omega}) = e^{-j\omega\frac{M-1}{2}} \left[h\left[\frac{M-1}{2}\right] + \sum_{n=0}^{(M-3)/2} 2h[n] \cos\left(\omega \left(\frac{M-1}{2} - n \right) \right) \right] \tag{71}$$

This explicitly shows that the frequency response has a purely real amplitude magnitude function multiplied by a perfectly linear phase shift term $e^{-j\omega(M-1)/2}$.

Question 2(b) (OR): 2-Dimensional FIR Offline Analysis

Problem Statement

Give a step-by-step procedure of employing two-dimensional discrete convolution summation for offline analysis of two-dimensional data using FIR digital filters.

1. Concept of Offline 2D Data Processing

In online (real-time) processing, systems are fundamentally restricted by the arrow of time: a filter cannot utilize future samples because they have not occurred yet. This mandates the use of **causal filters**.

However, in **offline analysis** (e.g., analyzing an existing digital image or pre-recorded geographical data), the entire dataset is resident in memory simultaneously. There is no concept of "future" time limiting the algorithm. Because of this, we can employ **non-causal, zero-phase FIR filters**. These filters are symmetrical around the origin ($n = 0$), eliminating the $(M - 1)/2$ time delay entirely and preventing any spatial shifting of the features in the data.

2. 2D Data Representation

A 2-dimensional dataset (such as a monochrome digital image) is represented as a 2D matrix of discrete samples $x[k, l]$, where:

- $k = 0, 1, \ldots, (M_x - 1)$ represents the rows (vertical axis).
- $l = 0, 1, \ldots, (N_y - 1)$ represents the columns (horizontal axis).

3. Defining the 2D FIR Filter Mask

For offline filtering, we design a 2-dimensional non-causal FIR filter mask $h[m, n]$. To maintain zero phase and perfect symmetry in both dimensions, the dimensions of the mask $p \times p$ are strictly chosen to be **odd integers**. The indices of the mask are symmetrically bounded around the origin $(0, 0)$:

$$-\frac{p-1}{2} \leq m \leq \frac{p-1}{2} \quad \text{and} \quad -\frac{p-1}{2} \leq n \leq \frac{p-1}{2} \tag{72}$$

For example, a 3×3 mask ($p = 3$) has indices running from -1 to $+1$.

4. The 2D Discrete Convolution Summation

The filtering procedure slides this $p \times p$ mask across every available pixel of the input data $x[k, l]$. The filtered output pixel $y[k, l]$ is computed by multiplying the overlapping elements and summing them up (the 2D convolution):

$$\boxed{y[k, l] = \sum_{m=-\frac{p-1}{2}}^{\frac{p-1}{2}} \sum_{n=-\frac{p-1}{2}}^{\frac{p-1}{2}} h[m, n] \cdot x[k - m, l - n] \tag{73}}$$

5. Step-by-Step Procedure & Boundary Limitations

1. **Center the Mask:** Place the origin $(0, 0)$ of the filter mask $h[m, n]$ directly over the target data pixel $x[k, l]$.

2. **Multiply and Accumulate:** Multiply each filter weight by the data value beneath it. Add all p^2 products to yield the output $y[k, l]$.

3. **Traverse the Grid:** Shift the mask right by one pixel and repeat. Once a row is complete, shift down to the next row.

4. **Boundary Truncation:** Because the mask extends outwards from the center pixel by $(p - 1)/2$ steps in every direction, we cannot calculate the convolution at the extreme edges of the image without "falling off" the data matrix. Therefore, the valid output indices for $y[k, l]$ are restricted. The output dataset is slightly smaller than the input dataset. The valid range is:

$$\frac{p-1}{2} \leq k \leq (M_x - 1) - \frac{p-1}{2} \tag{74}$$

$$\frac{p-1}{2} \leq l \leq (N_y - 1) - \frac{p-1}{2} \tag{75}$$

Question 3: Direct Realization Multipliers and Adders

Problem Statement

Show that the direct realization of an M-th order linear phase FIR digital filter can be achieved using $(M+1)/2$ number of multiplications, when M is odd. What will be the number of additions needed?

1. Standard FIR Direct Form Realization

An M-th order FIR filter difference equation utilizes M delay elements (z^{-1}) to hold past values. The length of the impulse response sequence is $M+1$ samples, ranging from h_0 to h_M. The standard generic difference equation is:

$$y[n] = \sum_{k=0}^{M} h_k x[n-k] = h_0 x[n] + h_1 x[n-1] + \cdots + h_M x[n-M] \quad (76)$$

In a direct (brute-force) implementation, calculating one output sample $y[n]$ directly requires exactly $M+1$ **separate multiplications** and M **additions**.

2. Exploiting Linear Phase Symmetry

We established in Question 2(b) that a causal FIR filter maintains strict linear phase if and only if its impulse response coefficients are symmetrically mirrored around the center. Mathematically, the symmetry constraint is:

$$h_k = h_{M-k} \quad \text{for } 0 \leq k \leq M \tag{77}$$

Let us assume M is an odd integer. Because M is odd, the number of coefficients (the filter length $L = M + 1$) is an **even number**. For an even number of coefficients, the sequence splits perfectly into two identical halves without a unique center coefficient. Let's expand the summation for an odd M (e.g., $M = 5$, yielding 6 terms h_0 to h_5):

$$y[n] = h_0x[n]+h_1x[n-1]+\cdots+h_{M-1}x[n-(M-1)]+h_Mx[n-M] \tag{78}$$

Applying the symmetry rule $h_M = h_0$, $h_{M-1} = h_1$, etc.:

$$y[n] = h_0x[n] + h_1x[n-1] + \cdots + h_1x[n-(M-1)] + h_0x[n-M] \tag{79}$$

We can group the terms that share the exact same multiplier coefficient h_k:

$$y[n] = h_0(x[n] + x[n-M]) + h_1(x[n-1] + x[n-(M-1)]) + \dots \tag{80}$$

In a generalized mathematical summation format, we fold the sequence exactly in half:

$$y[n] = \sum_{k=0}^{\frac{M-1}{2}} h_k \left(x[n-k] + x[n-(M-k)]\right) \tag{81}$$

3. Calculation of Multiplications and Additions

By examining the folded summation equation above:

- **Multiplications:** The summation index k runs from 0 up to $(M-1)/2$. The number of discrete terms in this summation is:

$$\frac{M-1}{2} - 0 + 1 = \frac{M-1+2}{2} = \frac{M+1}{2} \tag{82}$$

  Thus, by pre-adding the mirrored data samples together, we only need to multiply by the coefficient h_k once. **The total number of multiplications is strictly reduced to $(M+1)/2$.**

- **Additions:** The number of additions remains exactly the same as the non-symmetric case. We perform $(M+1)/2$ additions inside the parentheses to pair up the signals, and then we must add those $(M+1)/2$ scaled pairs together, requiring $((M+1)/2-1)$ additions. Total Additions $= \frac{M+1}{2} + \frac{M+1}{2} - 1 = \frac{2M+2}{2} - 1 = M + 1 - 1 = M$. **The total number of additions needed is M.**

This modified realization structure is known as the **Linear-Phase Direct Form** and operates by physically folding the tapped delay line back onto itself before routing to the multipliers, practically cutting the hardware DSP footprint in half.

Question 3 (OR): FIR Design & Gibbs Phenomenon

Problem Statement

Show that, in rectangular window based design of FIR filters, the frequency response of the filter designed can be expressed in form of a circular complex convolution integral. What is Gibbs phenomenon? How can it be reduced?

1. The Windowing Method and Circular Convolution

To design an ideal FIR digital filter, we start with an ideal frequency response $H_d(\omega)$ (e.g., a perfect "brick-wall" low-pass filter). We determine its ideal discrete-time impulse response $h_d[n]$ by taking the Inverse Discrete-Time Fourier Transform (IDTFT):

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \tag{83}$$

Because the ideal frequency response $H_d(\omega)$ contains sharp, discontinuous transitions, its inverse transform $h_d[n]$ exists as a mathematically infinite sequence (specifically, a non-causal sinc function extending from $n = -\infty$ to ∞).

To make this physically realizable as an FIR filter, we must truncate $h_d[n]$ to a finite length M. This truncation is mathematically identical to multiplying the infinite sequence $h_d[n]$ by a finite-length sequence called a **Window Function**, $w[n]$. The simplest is the Rectangular Window, which is 1 for $0 \leq n \leq M-1$ and 0 elsewhere.

$$h[n] = h_d[n] \cdot w[n] \tag{84}$$

According to the fundamental theorems of Fourier analysis, multiplication in the discrete time domain maps to periodic (circular) convolution in the frequency domain. Therefore, taking the DTFT of both sides yields the final frequency response of the realized filter $H(\omega)$:

$$\boxed{H(\omega) = \frac{1}{2\pi}\left[H_d(\omega) \circledast W(\omega)\right] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta)W(\omega-\theta)d\theta \qquad (85)}$$

where $W(\omega)$ is the frequency response (spectrum) of the rectangular window function.

2. What is the Gibbs Phenomenon?

The frequency response of a rectangular window $W(\omega)$ resembles a *Sinc* function; it possesses a narrow main lobe centered at $\omega = 0$ and infinite oscillating side lobes extending outwards.

When the ideal brick-wall spectrum $H_d(\omega)$ is circularly convolved with $W(\omega)$, the presence of these side lobes causes severe oscillatory behavior (ringing) in the final filter response $H(\omega)$ precisely near the edges of the passband and stopband transitions.

As the filter order M increases, the transition band becomes narrower (steeper), but the peak amplitude of the oscillation ripples **does not decrease**. The overshoot remains fixed at approximately **8.95%** of the discontinuity magnitude regardless of how large M gets. This inability to completely eliminate the ripples around abrupt discontinuities through simple truncation is universally known as the **Gibbs Phenomenon**.

3. How to Reduce the Gibbs Phenomenon

The Gibbs phenomenon is a direct consequence of the abrupt, sheer drop-off of the rectangular window in the time domain, which generates the massive side lobes in the frequency domain. To suppress the ringing, we replace the rectangular window with **tapered window functions** that smoothly transition to zero at the edges ($n = 0$ and $n = M-1$). Prominent tapered windows include:

- **Hann Window:** $w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{M-1}\right)$
- **Hamming Window:** $w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right)$
- **Blackman Window:** Uses higher order harmonics for extreme tapering.

These tapered windows drastically decrease the magnitude of the side lobes in $W(\omega)$ at the cost of slightly widening the main lobe. When convolved, the reduced side lobes result in a much smoother $H(\omega)$ with greatly suppressed Gibbs oscillations, ensuring high attenuation in the filter's stopband.

Question 4: Short Notes on FFT Concepts

Problem Statement

Write a short note on any one of the following: (We will solve both)

(i) Splitting an $(N/2)$-point DFT into two $(N/4)$-point DFTs.

(ii) Computation of 4-point FFT.

(i) Splitting an $(N/2)$-point DFT into two $(N/4)$-point DFTs

This concept lies at the very heart of the **Radix-2 Decimation-in-Time (DIT) Fast Fourier Transform (FFT) algorithm**. The central strategy of the FFT is "divide and conquer." The formula for an N-point DFT is:

$$X_k = \sum_{n=0}^{N-1} x_n W_N^{nk} \quad \text{for } k = 0, 1, \ldots, N-1 \tag{86}$$

Suppose we have a sequence of length $N' = N/2$. The algorithm proceeds by physically decimating (splitting) the discrete-time sequence x_n into two smaller sequences consisting of the **even-indexed** samples and the **odd-indexed** samples.

$$X_k = \sum_{n=0,2,4\ldots} x_n W_{N'}^{nk} + \sum_{n=1,3,5\ldots} x_n W_{N'}^{nk} \tag{87}$$

We define new index variables: Let $n = 2r$ for the even sequence and $n = 2r + 1$ for the odd sequence, where r steps from 0 to $(N'/2) - 1$. Since

$N'/2 = N/4$, this limits the new sums exactly to $N/4$ bounds.

$$X_k = \sum_{r=0}^{N/4-1} x_{2r} W_{N'}^{2rk} + \sum_{r=0}^{N/4-1} x_{2r+1} W_{N'}^{(2r+1)k} \tag{88}$$

We utilize the twiddle factor property: $W_{N'}^2 = e^{-j\frac{2\pi}{N'}\cdot 2} = e^{-j\frac{2\pi}{N'/2}} = W_{N'/2}$.

$$X_k = \sum_{r=0}^{N/4-1} x_{2r} W_{N/4}^{rk} + W_{N'}^{k} \sum_{r=0}^{N/4-1} x_{2r+1} W_{N/4}^{rk} \tag{89}$$

Notice that the two summation terms are now mathematically identical to two separate $(N/4)$-point DFTs. Let G_k be the $(N/4)$-point DFT of the even elements, and H_k be the $(N/4)$-point DFT of the odd elements.

$$X_k = G_k + W_{N'}^{k} H_k \quad \text{for } 0 \leq k < N' \tag{90}$$

By continuing this recursive binary splitting down to 2-point DFT basic blocks (Butterflies), the computational complexity collapses from $\mathcal{O}(N^2)$ complex multiplications to $\mathcal{O}(N \log_2 N)$, securing the immense speed of the FFT.

(ii) Computation of 4-point FFT

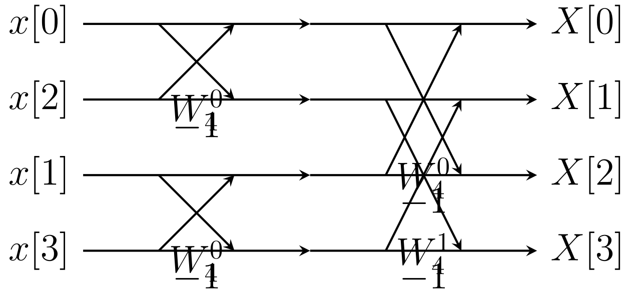
To compute a 4-point FFT ($N = 4$), we construct a signal flow graph consisting of $\log_2(4) = 2$ stages. The 4 input time-domain samples must first be rearranged into **Bit-Reversed Order**. For indexes $0, 1, 2, 3$ in binary (00, 01, 10, 11), bit reversal yields (00, 10, 01, 11), meaning the input sequence into the left side of the diagram is $x[0], x[2], x[1], x[3]$.

Twiddle Factors (W_N^k): For $N = 4$, the required twiddle factors are:

- $W_4^0 = e^0 = 1$

- $W_4^1 = e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}} = -j$

Signal Flow Graph (Radix-2 DIT Butterfly): The architecture relies on the basic DIT Butterfly, mapping inputs a, b to outputs A, B via $A = a + W^k b$ and $B = a - W^k b$.



Equations Evaluated: *Stage 1 Outputs:*

$$v_0 = x[0] + W_4^0 x[2]$$
$$v_1 = x[0] - W_4^0 x[2]$$
$$v_2 = x[1] + W_4^0 x[3]$$
$$v_3 = x[1] - W_4^0 x[3]$$

Final Stage 2 Outputs (Frequency Domain):

$$X[0] = v_0 + W_4^0 v_2$$
$$X[1] = v_1 + W_4^1 v_3$$
$$X[2] = v_0 - W_4^0 v_2$$
$$X[3] = v_1 - W_4^1 v_3$$

Question 5: FIR Filter Design (Fourier Series Method)

Problem Statement

A 5-tap FIR causal digital filter with stepped characteristic is designed. $H(\omega) = 1.1e^{-j\omega\tau(M-1)/2}$ for $|\omega| \leq \omega_1$, and $0.9e^{-j\omega\tau(M-1)/2}$ for $\omega_1 < |\omega| \leq \omega_2$. $f_1 = 200$ Hz, $f_2 = 400$ Hz, $f_s = 1000$ Hz. Smoothened using Hann window. Realize filter and draw schematic form.

1. Extracting Discrete Frequencies

First, we must normalize the continuous analog frequencies to discrete digital angular frequencies ($\omega = 2\pi f/f_s$):

- Sampling Frequency $f_s = 1000$ Hz $\implies \tau = 1/1000$ sec.
- Cutoff 1: $\omega_1 = 2\pi\left(\frac{200}{1000}\right) = 0.4\pi$ rad/sample.
- Cutoff 2: $\omega_2 = 2\pi\left(\frac{400}{1000}\right) = 0.8\pi$ rad/sample.

The filter order is $M = 5$ taps (length = 5). The phase delay constant is $\alpha = (M-1)/2 = 4/2 = 2$. The desired ideal causal frequency response is:

$$H_d(\omega) = \begin{cases} 1.1e^{-j2\omega} & 0 \leq |\omega| \leq 0.4\pi \\ 0.9e^{-j2\omega} & 0.4\pi < |\omega| \leq 0.8\pi \\ 0 & 0.8\pi < |\omega| \leq \pi \end{cases} \tag{91}$$

2. Evaluating the Ideal Non-Causal Impulse Response

Instead of dealing with the complex exponentials immediately, it is mathematically simpler to compute the non-causal symmetric impulse response $\hat{h}_d[n]$ centered at $n = 0$ (by temporarily ignoring the $e^{-j2\omega}$ phase delay), and then simply shifting the final answer right by $\alpha = 2$ steps to enforce causality.

The Inverse DTFT for the zero-phase magnitude function is:

$$\hat{h}_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} |H_d(\omega)|e^{j\omega n}d\omega = \frac{1}{\pi}\int_{0}^{\pi} |H_d(\omega)|\cos(\omega n)d\omega \tag{92}$$

Breaking the integral into the two stepped frequency bands:

$$\hat{h}_d[n] = \frac{1}{\pi}\left[\int_{0}^{0.4\pi} 1.1\cos(\omega n)d\omega + \int_{0.4\pi}^{0.8\pi} 0.9\cos(\omega n)d\omega\right] \tag{93}$$

For the center sample ($n = 0$):

$$\hat{h}_d[0] = \frac{1}{\pi}\left[\int_{0}^{0.4\pi} 1.1d\omega + \int_{0.4\pi}^{0.8\pi} 0.9d\omega\right] \tag{94}$$

$$\hat{h}_d[0] = \frac{1}{\pi}[1.1(0.4\pi - 0) + 0.9(0.8\pi - 0.4\pi)] = \frac{1}{\pi}[0.44\pi + 0.36\pi] = 0.800 \tag{95}$$

For $n = \pm 1$:

$$\hat{h}_d[\pm 1] = \frac{1}{\pi}\left[1.1\,\frac{\sin(\omega)}{1}\bigg|_{0}^{0.4\pi} + 0.9\,\frac{\sin(\omega)}{1}\bigg|_{0.4\pi}^{0.8\pi}\right] \tag{96}$$

$$\hat{h}_d[\pm 1] = \frac{1}{\pi}\left[1.1\sin(0.4\pi) + 0.9\sin(0.8\pi) - 0.9\sin(0.4\pi)\right] \tag{97}$$

$$\hat{h}_d[\pm 1] = \frac{1}{\pi}\left[0.2\sin(0.4\pi) + 0.9\sin(0.8\pi)\right] \tag{98}$$

Using $\sin(0.4\pi) = \sin(72^\circ) \approx 0.9511$ and $\sin(0.8\pi) = \sin(144^\circ) \approx 0.5878$:

$$\hat{h}_d[\pm 1] = \frac{1}{\pi}\left[0.2(0.9511) + 0.9(0.5878)\right] = \frac{0.1902 + 0.5290}{3.14159} = \frac{0.7192}{\pi} \approx 0.2289 \tag{99}$$

For $n = \pm 2$:

$$\hat{h}_d[\pm 2] = \frac{1}{2\pi}\left[1.1\sin(0.8\pi) + 0.9\sin(1.6\pi) - 0.9\sin(0.8\pi)\right] \tag{100}$$

$$\hat{h}_d[\pm 2] = \frac{1}{2\pi}\left[0.2\sin(0.8\pi) + 0.9\sin(1.6\pi)\right] \tag{101}$$

Using $\sin(1.6\pi) = \sin(288^\circ) \approx -0.9511$:

$$\hat{h}_d[\pm 2] = \frac{1}{2\pi}\left[0.2(0.5878) + 0.9(-0.9511)\right] = \frac{0.1176 - 0.8560}{2\pi} = \frac{-0.7384}{2\pi} \approx -0.1175 \tag{102}$$

3. Applying the Hann Window

To prevent Gibbs phenomenon, we multiply the ideal sequence by the Hann window. For a non-causal sequence of length $M = 5$, the window is centered at $n = 0$:

$$w_{hann}[n] = 0.5 + 0.5\cos\left(\frac{2\pi n}{M-1}\right) = 0.5 + 0.5\cos\left(\frac{\pi n}{2}\right) \quad \text{for } -2 \leq n \leq 2 \tag{103}$$

Evaluating the window weights:

- $w[0] = 0.5 + 0.5\cos(0) = 1.0$
- $w[\pm 1] = 0.5 + 0.5\cos(\pi/2) = 0.5$
- $w[\pm 2] = 0.5 + 0.5\cos(\pi) = 0.0$

Multiplying the ideal sequence by the window weights ($\hat{h}[n] = \hat{h}_d[n] \cdot w[n]$):

- $\hat{h}[0] = 0.8000 \times 1.0 = 0.8000$
- $\hat{h}[\pm 1] = 0.2289 \times 0.5 \approx 0.1145$
- $\hat{h}[\pm 2] = -0.1175 \times 0.0 = 0.0000$

4. Final Causal Filter Sequence and Realization

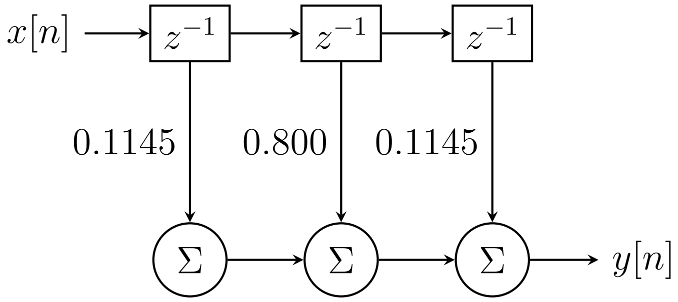
To make the filter physically causal, we reinstate the phase delay by shifting the non-causal sequence completely to the right by $(M-1)/2 = 2$ indices ($h[n] = \hat{h}[n-2]$).

$$h[0] = 0, \quad h[1] = 0.1145, \quad h[2] = 0.8000, \quad h[3] = 0.1145, \quad h[4] = 0 \tag{104}$$

Remarkably, because the outermost coefficients $h[0]$ and $h[4]$ have been completely forced to zero by the Hann window, this effectively operates as a 3-tap filter internally. The difference equation is:

$$y[n] = 0.1145x[n-1] + 0.8x[n-2] + 0.1145x[n-3] \tag{105}$$

Schematic Form Realization:



Question 5 (OR): Frequency Response of Blackman Window

Problem Statement

Derive the frequency response of a non-causal Blackman window...

Comment on its approximate width of the main lobe.

1. Derivation of the Frequency Response

The discrete time-domain sequence of a non-causal Blackman window of length M is defined symmetrically around the origin:

$$w[n] = 0.42 + 0.5\cos\left(\frac{2\pi n}{M-1}\right) + 0.08\cos\left(\frac{4\pi n}{M-1}\right) \quad \text{for } |n| \leq \frac{M-1}{2} \tag{106}$$

To find the frequency response $W(\omega)$, we take the Discrete-Time Fourier Transform (DTFT):

$$W(\omega) = \sum_{n=-(M-1)/2}^{(M-1)/2} w[n]e^{-j\omega n} \tag{107}$$

Using Euler's identity to expand the cosine terms into complex exponentials:

$$\cos\left(\frac{2\pi n}{M-1}\right) = \frac{1}{2}e^{j\frac{2\pi n}{M-1}} + \frac{1}{2}e^{-j\frac{2\pi n}{M-1}} \tag{108}$$

$$\cos\left(\frac{4\pi n}{M-1}\right) = \frac{1}{2}e^{j\frac{4\pi n}{M-1}} + \frac{1}{2}e^{-j\frac{4\pi n}{M-1}} \tag{109}$$

Substituting these into the $w[n]$ expression gives 5 distinct complex exponential terms:

$$w[n] = 0.42 + 0.25e^{j\frac{2\pi n}{M-1}} + 0.25e^{-j\frac{2\pi n}{M-1}} + 0.04e^{j\frac{4\pi n}{M-1}} + 0.04e^{-j\frac{4\pi n}{M-1}} \tag{110}$$

Recall that the DTFT of a simple uniform non-causal Rectangular Window $w_R[n] = 1$ is a discrete Sinc function:

$$W_R(\omega) = \frac{\sin(\omega M/2)}{\sin(\omega/2)} \tag{111}$$

By the frequency shifting property of the Fourier Transform ($\mathcal{F}[x[n]e^{j\omega_0 n}] = X(\omega - \omega_0)$), each exponential term in $w[n]$ acts to shift the rectangular response $W_R(\omega)$ across the frequency axis. Thus, the total frequency response $W(\omega)$ is a linear combination of shifted Sinc functions:

$$\begin{aligned} W(\omega) = \quad & 0.42W_R(\omega) \\ & + 0.25\left[W_R\left(\omega - \frac{2\pi}{M-1}\right) + W_R\left(\omega + \frac{2\pi}{M-1}\right)\right] \\ & + 0.04\left[W_R\left(\omega - \frac{4\pi}{M-1}\right) + W_R\left(\omega + \frac{4\pi}{M-1}\right)\right] \end{aligned} \tag{112}$$

2. Approximate Width of the Main Lobe

The central rectangular component $W_R(\omega)$ has its first zero crossings at $\omega = \pm 2\pi/M$, giving it a main lobe width of $4\pi/M$. The Blackman window incorporates secondary shifts at $\pm 2\pi/(M-1)$ and tertiary shifts at $\pm 4\pi/(M-1)$. The superposition of these highly shifted Sinc functions forces the primary nulls of the final $W(\omega)$ response outwards. The first combined zero crossing occurs at approximately $\pm 6\pi/M$. Therefore, the approximate width of the main lobe is:

$$\boxed{\text{Main Lobe Width} \approx 2 \times \frac{6\pi}{M} = \frac{12\pi}{M} \qquad (113)}$$

This demonstrates that the Blackman window possesses an extraordinarily wide main lobe (three times wider than a rectangular window), but in return offers unprecedented side-lobe attenuation (nearly -74 dB), minimizing Gibbs ripples.

Question 6: Justification of Statements

Problem Statement

Justify or correct any two of the following statements with suitable reasons/derivations, in brief. (We will verify all three).

(i) DFT Coefficients relation to Fourier Series

Statement: The DFT coefficients X_n for an N-point sequence can be obtained from a_n and b_n by $X_n = (\frac{N}{2})(a_n + jb_n)$.

Status: INCORRECT.

Correction & Justification: The continuous periodic trigonometric Fourier Series is defined with real coefficients a_n and b_n. Its relationship to the complex Fourier coefficient c_n is:

$$c_n = \frac{1}{2}(a_n - jb_n) \tag{114}$$

When a continuous signal is digitized, the Discrete Fourier Transform (DFT) evaluates the spectrum. The standard DFT definition operates without the $1/N$ normalizing factor on the forward transform ($X_k = \sum x_n W_N^{nk}$). Because of this, the magnitude of the DFT coefficients scales directly with N compared to the continuous coefficients:

$$X_n = N \cdot c_n = N\left(\frac{1}{2}(a_n - jb_n)\right) = \frac{N}{2}(a_n - jb_n) \tag{115}$$

The provided statement incorrectly uses a positive sign for the imaginary component $+jb_n$. Due to the $e^{-j\omega t}$ term in the forward transform containing $\cos(\theta) - j\sin(\theta)$, the relation strictly carries a negative sign.

(ii) Radix-2 DIF FFT "In-Place" Feature

Statement: Radix-2 decimation-in-frequency FFT algorithm can incorporate an "in-place" feature because of the structure of each butterfly employed.

Status: CORRECT.

Justification: The basic computation engine of the FFT is the Butterfly diagram. A single Radix-2 butterfly block takes exactly two complex input nodes in memory (say, array elements $x[A]$ and $x[B]$) and computes exactly two complex output nodes ($X[A]$ and $X[B]$) using the formulas $X[A] = x[A] + x[B]$ and $X[B] = (x[A] - x[B])W_N^k$. Crucially, once these two outputs are computed, the original input values $x[A]$ and $x[B]$ are never needed again for any subsequent stage of the algorithm. Therefore, the outputs $X[A]$ and $X[B]$ can be immediately written directly back into the exact same memory addresses that held the inputs. This eliminates the need for auxiliary memory arrays, permitting the FFT to compute completely **"in-place"** with $\mathcal{O}(1)$ auxiliary memory requirement.

(iii) Phase Delay vs Group Delay for Distortion-less Transmission

Statement: Both phase delay and group delay can be equally employed in all conditions to test whether a digital filter permits distortion-less transmission.

Status: INCORRECT.

Correction & Justification: For a filter to transmit a signal without shape distortion, all frequency components must be delayed by the exact same amount of time in seconds. Phase delay is defined as $\tau_p(\omega) = -\frac{\theta(\omega)}{\omega}$, and Group delay is $\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega}$. If a filter has a phase response $\theta(\omega) = -k\omega$ (strictly linear crossing the origin), then $\tau_p = k$ and $\tau_g = k$. Both are constant, and both successfully indicate distortionless transmission. However, if a filter has a phase response with a fixed intercept, such as $\theta(\omega) = -k\omega + \beta$, the group delay evaluates to a constant k, but the phase delay evaluates to $\tau_p = k - \beta/\omega$, which is highly frequency-dependent. Because group delay is a derivative, it masks constant phase shifts. Therefore, constant group delay alone does NOT guarantee true distortionless transmission if phase intercept shifts are present. **Constant Phase Delay** is the stricter, definitive condition.